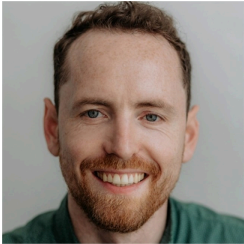


Dr. John D. Kirwan

Computational Biologist · Bioinformatics & Machine Learning

body



KEY SKILLS

- Python
- R
- PyTorch
- Machine Learning
- Deep Learning
- Bayesian Statistics
- Protein Structure Prediction
- Structural Bioinformatics
- Metagenomics
- Docker

PROFILES

GitHub: JohnKirwan
LinkedIn: johndkirwan
ORCID: 0000-0001-5537-3574

LANGUAGES

English (Native)
Spanish (B2)
Swedish (B2)

PROFILE

Computational biologist and data scientist pairing rigorous statistical modelling — Bayesian inference and deep learning — with hands-on ML engineering and reproducible pipelines. Nearly four years at the interface of biotech and AI in an industry research setting, spanning biomolecular structure prediction, generative molecular design, and metagenomic analysis, and including a contribution to a NeurIPS 2025 publication on uncertainty quantification. Now focusing on structure-informed metagenomics, functional annotation, and antimicrobial-resistance research — applying structural and evolutionary insight to large-scale biological data.

SELECTED HIGHLIGHTS

- Computational biologist with nearly four years at the AI-biotech interface in industry research
- Experience spanning structure prediction, binding-affinity modelling, metagenomics, and uncertainty quantification
- Hands-on ML engineering with PyTorch, Docker, AWS, Azure DevOps, and GPU workflows
- Research background grounded in Bayesian modelling, quantitative biology, and reproducible analysis

RECENT EXPERIENCE

Bioinformatics Scientist *Arquimea Research Centre, La Laguna, Spain* (Mon Jul 04 2022 01:00:00 GMT+0100 (Western European Summer Time)-Wed Mar 11 2026 00:00:00 GMT+0000 (Western European Standard Time)) Worked at the interface of the AI and biotech teams on structure-based drug discovery, generative molecular design, and applied machine learning — the link between the computational tools and the biology. Contributing author on a NeurIPS 2025 publication on uncertainty quantification in deep regression.

- Built a modular synthetic-data-generation workflow orchestrating GNINA, DiffDock and AlphaFold 2; sourced datasets and handled inter-tool format conversion
- Predicted protein structures and binding affinities for protein complexes, including metal-ion and ATP cofactors, using Boltz, AlphaFold, Foldseek and BioEMU
- On the engine team of Uncharted, an internal venture building a multi-objective small-molecule optimisation platform; developed the binding-affinity objective

Postdoctoral Researcher *Stazione Zoologica Anton Dohrn, Naples, Italy* (Tue Oct 01 2019 01:00:00 GMT+0100 (Western European Summer Time)-Fri Jul 22 2022 01:00:00 GMT+0100 (Western European Summer Time)) Quantitative sensory biology within an international Human Frontier Science Program (HFSP) collaboration, with emphasis on statistical modelling and signal processing of behavioural data.

- Analysed experimental data using Bayesian multilevel regression models in R and Stan
- Designed and executed behavioural experiments with automated data acquisition pipelines
- Contributed to interdisciplinary publications spanning biology, optics, and computation

EDUCATION

PhD Sensory Biology *Lund University, Sweden* (Wed May 01 2013 01:00:00 GMT+0100 (Western European Summer Time)-Mon May 07 2018 01:00:00

GMT+0100 (Western European Summer Time)) Thesis: 'Spatial Vision in Diverse Invertebrates'.

MSc Molecular Evolution *University College Dublin, Ireland* (Tue Dec 09 2008 00:00:00 GMT+0000 (Western European Standard Time)-Tue Jun 01 2010 01:00:00 GMT+0100 (Western European Summer Time)) Thesis: 'The Molecular Evolution of Hearing in Mammals'

ADDITIONAL EXPERIENCE

Project Manager (contract) *Lund University, Lund, Sweden* (Sat Jun 01 2019 01:00:00 GMT+0100 (Western European Summer Time)-Sat Aug 31 2019 01:00:00 GMT+0100 (Western European Summer Time)) Short-term research management contract supervising experimental data collection and a masters student, bridging doctoral and postdoctoral positions.

Doctoral Researcher *Lund University, Lund, Sweden* (Wed May 15 2013 01:00:00 GMT+0100 (Western European Summer Time)-Thu Jun 07 2018 01:00:00 GMT+0100 (Western European Summer Time)) Computational and statistical analysis of visual systems in invertebrates, combining experimental biology with quantitative modelling.

- Built statistical models using Bayesian multilevel regression in R and Stan; additional analysis in Python and Matlab
- Developed optical and geometric models to estimate spatial resolution theoretically
- Designed novel experimental protocols for quantifying animal visual performance

ADDITIONAL EDUCATION

BSc (Hons) Zoology *University College Dublin, Ireland* (Sun Sep 12 2004 01:00:00 GMT+0100 (Western European Summer Time)-Sat May 31 2008 01:00:00 GMT+0100 (Western European Summer Time))

- First Class Honours (GPA: 4.02/4.2)

AWARDS & CERTIFICATIONS

- Pharmaceutical Bioinformatics & Applied Pharmaceutical Bioinformatics — Uppsala University — Sun May 31 2020 01:00:00 GMT+0100 (Western European Summer Time) Pass with Distinction
- Customising your models with TensorFlow 2 — Coursera / Imperial College London — Thu Aug 14 2025 01:00:00 GMT+0100 (Western European Summer Time)
- Valohai MLOps Fundamentals Certification — Valohai — Tue Aug 12 2025 01:00:00 GMT+0100 (Western European Summer Time)
- Applied Software Engineering Fundamentals Specialization — Coursera / IBM — Fri Mar 10 2023 00:00:00 GMT+0000 (Western European Standard Time) Covers Git, GitHub, Linux, and Python, and the fundamentals of software engineering.
- Mountain Leader Award — Mountain Leader Training Scotland — Sat Oct 01 2011 01:00:00 GMT+0100 (Western European Summer Time) Professional certification to lead hillwalking groups.

RESEARCH FOCUS

Computational biologist and data scientist with a track record of applying machine learning and statistical modelling to complex biological problems — from biomolecular structure prediction and de novo peptide design to metagenomic analysis and preclinical trial design.

My background combines rigorous quantitative training (Bayesian modelling, deep learning, MLOps) with hands-on experience across the full data pipeline — from experimental design through to deployment and reproducible reporting. I have worked at the interface of Biotech and AI in an industry research setting, contributing to work that reached commercial spinout stage and to a NeurIPS 2025 publication on uncertainty quantification in deep regression.

I am now focusing on structure-informed metagenomics, functional annotation, and antimicrobial-resistance research — applying the same analytical rigour to questions where structural and evolutionary insight meets large-scale biological data, with an eye to applications across health, agriculture, and the environment.